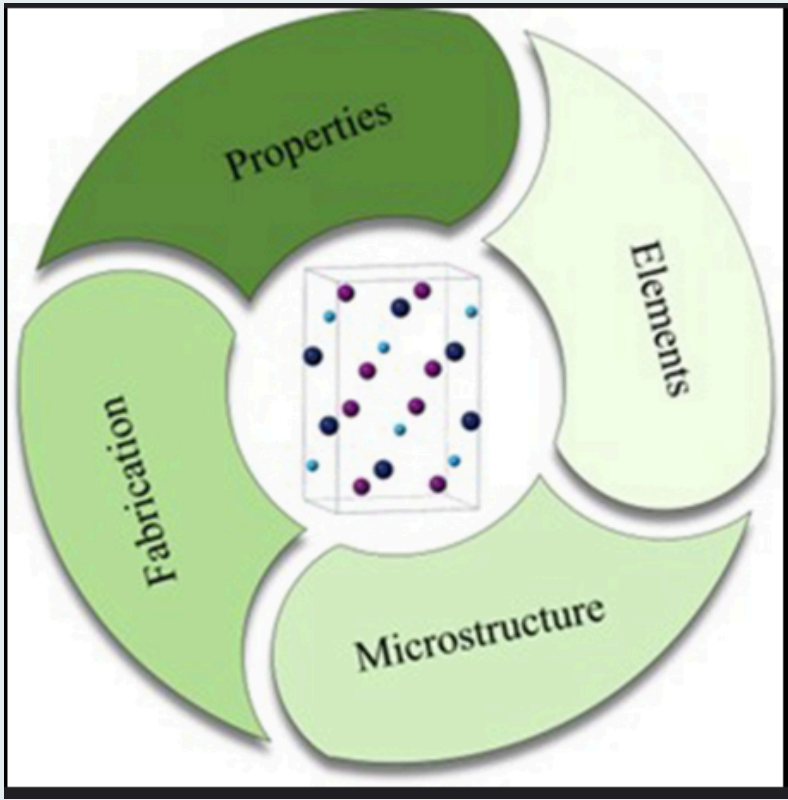


MICROSTRUCTURAL ANALYSIS OF BINARY ALLOYS FOR DIFFERENT PROCESSING(BINARY AL-ZN ALLOYS FABRICATED BY CASTING AND HEAT TREATMENT)

Binary alloys, composed of two elements, are key in materials science for their customizable properties, ideal for aerospace, automotive, and construction. Al-Zn alloys stand out for their lightweight, high strength, and corrosion resistance, enhanced by zinc's solid-solution strengthening. Tailorable via heat treatment, they excel in structural components and coatings, driving innovation in durable, sustainable technologies.



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INTRODUCTION

Binary Al-Zn alloys with varying Zn contents were fabricated through casting and heat treatment to study their microstructures and mechanical properties. The research investigates the effects of Zn content and solid-solution treatment on hardness, tensile strength, and impact toughness.

OBJECTIVE

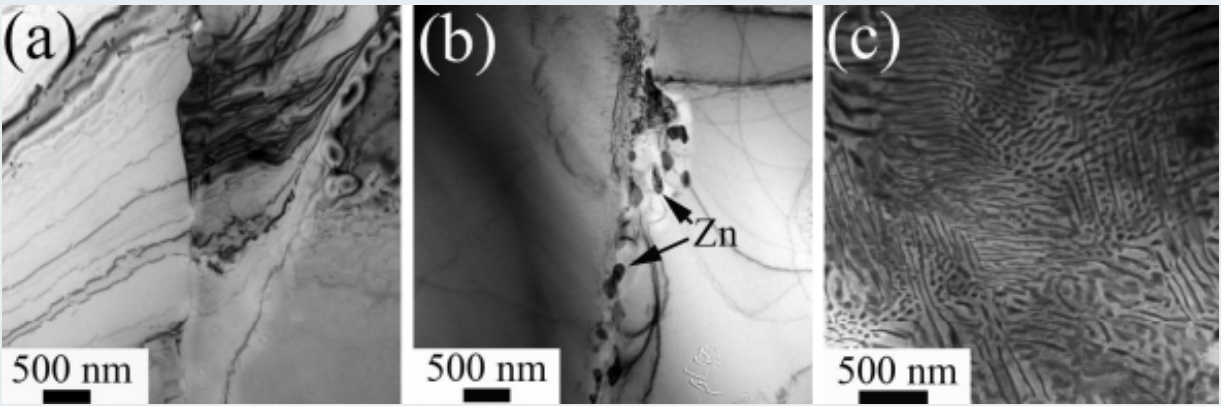
To investigate the effects of Zn content and solid-solution treatment on the microstructures and mechanical properties of binary Al-Zn alloys, focusing on hardness, tensile strength, and impact toughness, to optimize their performance for engineering applications.

METHODOLOGY

- Alloys prepared: Al-20Zn, Al-35Zn, Al-49Zn
- Processes: Casting followed by solid-solution treatment at 673–803 K for 5 hours
- Analysis:
 - X-ray diffraction (XRD) for phase confirmation
 - Vickers microhardness (3000 gf, 15 s)
 - Tensile tests (Instron 5982, strain rate $1 \times 10^{-4} \text{ s}^{-1}$)
 - Charpy impact tests (2 mm V-notch samples)
 - SEM, TEM, and EDS for microstructure analysis

RESULTS: MICROSTRUCTURE

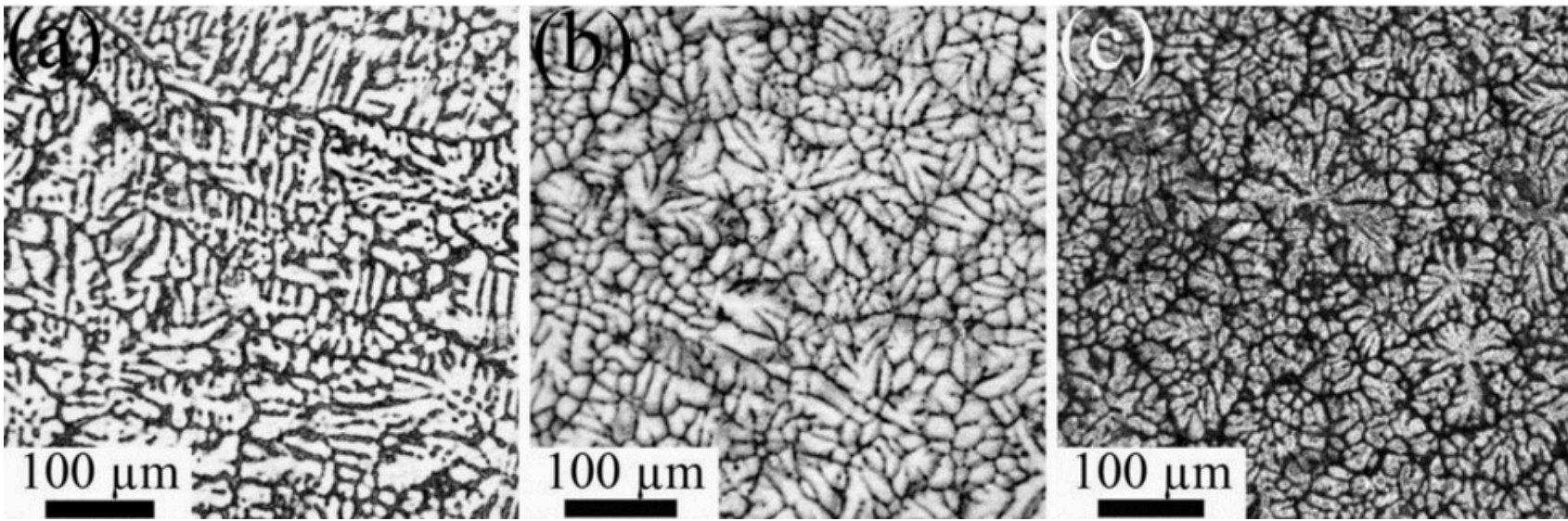
- As-Cast: Dendritic structure with reduced arm spacing as Zn content increased (e.g., 20 μm in Al-49Zn).
- After Solid-Solution (SS): Dendritic grains transformed to equiaxed grains ($\sim 200 \mu\text{m}$). Zn segregation eliminated, with refined Zn phase in Al-49Zn.
- TEM: Nanoscale Zn lamellae in as-cast Al-49Zn; smaller, uniform Zn phase in as-SS Al-49Zn.
- XRD: Al peaks shifted to high-angle post-treatment, indicating solid-solution reaction



TEM IMAGES OF (A) AS-SS AL-20ZN ALLOY TREATED AT 530 C FOR 5 H, (B) AL-35ZN ALLOY TREATED AT 470 C FOR 5 H, AND (C) AL-49ZN ALLOY TREATED AT 400 C FOR 5 H

RESULTS: MECHANICAL PROPERTIES

- Hardness: Increased with Zn content; as-SS Al-49Zn reached 152 HV due to solid-solution and Orowan strengthening.
- Tensile Properties: Strength increased with Zn, but elongation decreased. Al-20Zn showed 6.8% elongation; Al-49Zn had 330 MPa ultimate strength.
- Fracture Mode: Ductile fractures in Al-20Zn and Al-35Zn; brittle cleavage in Al-49Zn.
- Impact Toughness: Decreased with Zn content; Al-20Zn showed 58 kJ/m^2 , comparable to Al-Si alloys.



OM images of as-cast (a) Al-20Zn, (b) Al-35Zn, and (c) Al-49Zn alloys

CONCLUSION

- As-cast Al-Zn alloys exhibited dendritic structures with Zn segregation, refined by solid-solution treatment.
- Hardness and tensile strength increased with Zn content, with as-SS Al-49Zn showing superior hardness (152 HV).
- Impact toughness decreased with higher Zn content; Al-20Zn offered balanced strength and ductility.
- Solid-solution treatment enhanced mechanical properties via uniform Zn distribution and phase refinement.

Reference

W.B. Zhou, G.B. Teng, C.Y. Liu, H.Q. Qi, H.F. Huang, Y. Chen, and H.J. Jiang JMEPEG (2017) 26:3977–3982 ASM International DOI: 10.1007/s11665-017-2852-y Microstructures and Mechanical Properties of Binary Al-Zn Alloys Fabricated by Casting and Heat Treatment (Submitted November 14, 2016; in revised form June 6, 2017; published online August 4, 2017)

Link For the Paper:<https://link.springer.com/article/10.1007/s11665-023-08254-0>

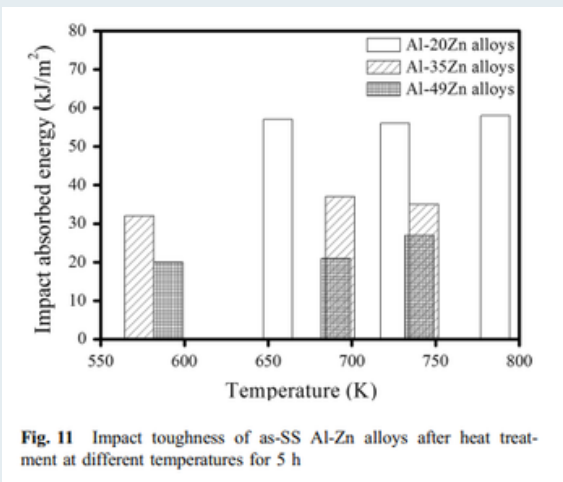


Fig. 11 Impact toughness of as-SS Al-Zn alloys after heat treatment at different temperatures for 5 h

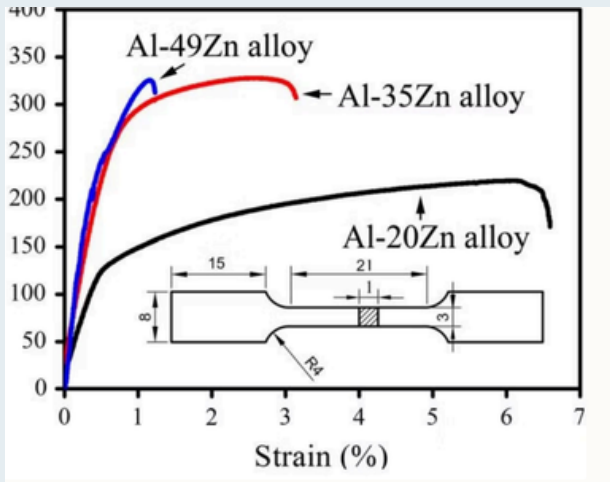
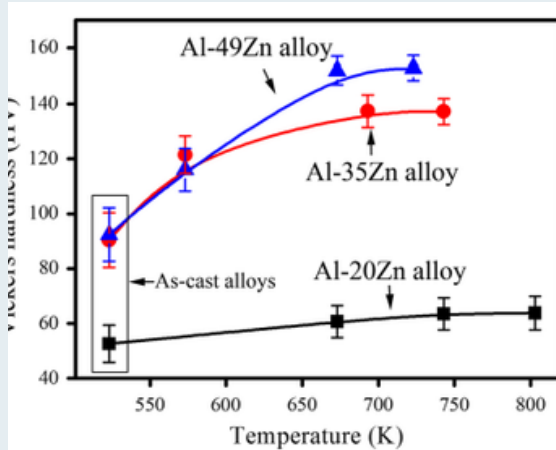


Fig. 7 XRD of as-cast Al-49Zn alloy, as-SS Al-20Zn, and Al-49Zn



g. 8 Vickers hardness of as-cast and as-SS Al-Zn alloys after treatment at different temperatures for 5 h